

Q2

Known

$L_A = 30 \text{ cm}$, $\dot{q}$ heat gen, k_A

left side thermally insulated

$L_B = 10 \text{ cm}$ $k_B = 30 \text{ W/mK}$

$T_\infty = 20^\circ\text{C}$ $h = 500 \text{ W/m}^2\text{K}$

$T_w = 115^\circ$ $T_i = 100^\circ\text{C}$

Find

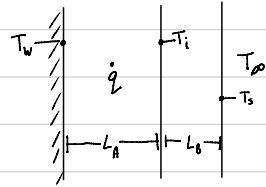
a. right surface temp of B (T_s)

b. $\dot{q}$ @ s.s.

c. k_A of A

d. Plot T profiles of A & B

Schematic



properties

Steady state, conduction, convection

assumptions

- perfectly insulated on left side
- steady state

analysis

a. $\dot{q}_{in} = \dot{q}_{out}$

$$-k \frac{dT}{dx} = h(T_s - T_\infty) \quad \rightarrow \quad -k \frac{T_s - T_i}{L_B} = h(T_s - T_\infty)$$

$$30 \frac{(100 - T_s)}{0.1} = (500)(T_s - 20)$$

$$30000 - 300T_s = 500T_s - 10000$$

$$40000 = 800T_s \rightarrow \boxed{T_s = 50^\circ\text{C}}$$

b. $\dot{q}_{gen} = -k \frac{dT}{dx}$

$$= 30 \frac{(100 - 50)}{0.1} = 15000$$

$(.3) \dot{q} = 15000 \quad \dot{q} = 50000 \text{ W/m}^3 \quad \boxed{\dot{q}_{gen} = 50 \text{ kW/m}^3}$

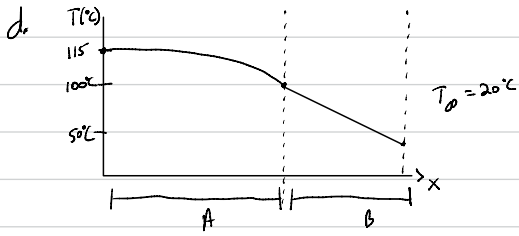
c. $T = \frac{-\dot{q}}{2k} x^2 + \frac{\dot{q}}{h} + \frac{\dot{q} L^2}{2k} + T_\infty$

$$100 = \frac{-(50000)}{2k} (.3)^2 + 115$$

$$-15 = \frac{-(50000)}{2k} (.3)^2 \rightarrow 2k = \frac{(50000)(.3)^2}{15}$$

$$k = \frac{(50000)(.3)^2}{2(15)} = 150$$

$$\boxed{k_A = 150 \text{ W/m.K}}$$



The temperature drops from 50°C to 20°C
From the edge of wall B to the ambient
environment